Python Programming - Study Notes

SmartCampus Cloud | Sample Student Resource

1. Introduction

Python is a high-level, general-purpose programming language known for readable syntax. It is used in web development, automation, data analysis, and machine learning.

2. Variables and Data Types

A variable stores a value. Common built-in types include int (whole numbers), float (decimal numbers), str (text), and bool (True or False).

3. Conditions and Loops

Use if, elif, and else to make decisions. Use for and while loops to repeat a block of code.

4. Functions

Define a function with def. Functions help organize reusable code and may accept parameters and return results.

5. Example

```
def greet(name):
return f'Hello, {name}'

print(greet('Student'))
```

Revision Questions

1. What is a variable?
2. Explain the difference between a for loop and a while loop.
3. What is the purpose of a function?